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Decay Effects in Online Advertising: Quantifying the Impact of Time Since Last Exposure

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Starcom IP
Dynamic Logic

Starcom IP/Dynamic Logic Partnership

- **Starcom IP** and **Dynamic Logic** have collaborated on more than 50 advertising effectiveness studies measuring the effects of both Internet and Television advertising

- Starcom IP
  - Leading interactive agency

- Dynamic Logic
  - Independent research company specializing in measuring marketing effectiveness
  - Has conducted studies for 47 out of Top 50 U.S. Advertisers

Purpose of this research

- To quantify the decay dynamics of brand metrics following OTS of online advertising
  - Magnitude of decay
  - Rate of decay
- Brand metrics of interest include:
  - Aided Brand Awareness
  - Aided Online Advertising Awareness
  - Message Association
  - Brand Favorability
  - Purchase Intent
- Outcome will inform:
  - Media planning decisions such as scheduling
  - Impact of decay on Internet-based ad effectiveness measurement

Research hypothesis

- Expectation is that as time since last exposure increases, branding scores will decrease
- Because minimal literature exists, analysis was a discovery process designed to uncover:
  - Rate of decay (How fast the decay occurs)
  - Nature of decay (Which metrics are most affected)

Approaches explored

- Primary data
  - Specifically designed to measure decay
  - One brand in relatively static situation
  - Apply AdIndex methodology to capture exposed respondents at some point following OTS
  - Benefits: controlled environment, homogenous data
  - Cons: cost and timing; limited data set
- Normative data
  - Mine Dynamic Logic MarketNorms database
  - Based on time stamp field, identify cases which represent various periods of times between OTS and survey completion
  - Benefits: large amount of data readily available, cost, timing
  - Cons: Heterogeneous data could limit analysis

Data summary

- Stratified sample of 2003 MarketNorms data
  - All cases where difference between OTS and survey completion is greater than one day
  - Random sample of 5,200 cases where survey occurred immediately following OTS
- Respondents gathered from surveys across CPG, Auto, and Pharmaceutical verticals
  - Total sample n=28,416
- Models contained multiple controls, including frequency

MarketNorms overview

- A vast normative database designed to be used as a strategic research, decision support and comparative tool for measuring online advertising and marketing effectiveness
  - 4+ years of data
  - 1200+ campaigns
  - 14,000+ creatives
  - 1,200,000+ respondents
- Data collected using a control/exposed research design which isolates the effects of online advertising from other variables; most data collected shortly after last ad exposure
- Variables measured that are relevant to the study:
  - Time since last exposure (electronic data)
  - Frequency (electronic data)
  - Demographics (survey data)
  - Industry category (study classification)

Analytical approaches

- Respondent-level data
  - Logistic regression
  - Independent variables: Frequency and recency of OTS, age, gender, income
  - Dependent variables: Binomial indication of awareness or persuasion

- Aggregated-level data
  - Cross-tab brand metrics with time (in days) since exposure
  - OLS regression against the tabbed data
  - Independent variable: Recency of OTS
  - Dependent variable: Percent of respondents at Day X who indicate affirmative response to brand metric

Results: Respondent Level Data

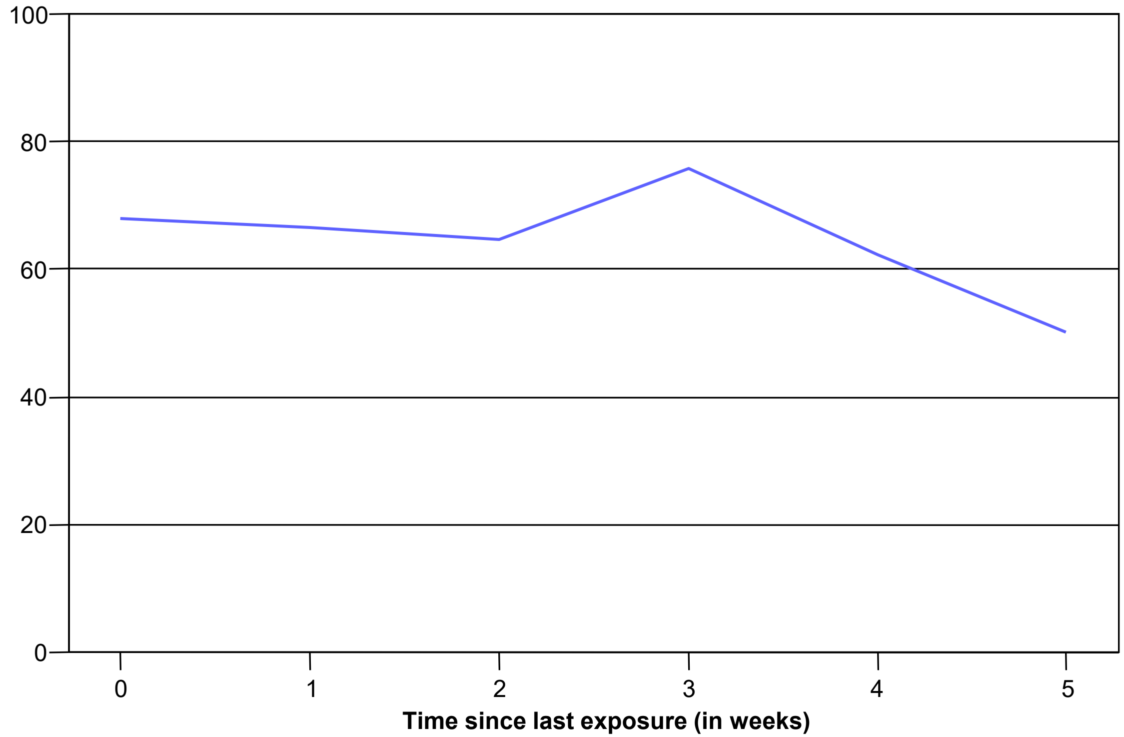
Findings – Respondent-level

- Binary logistic regression testing relationship between time since last exposure (day intervals) and brand scores
  - Data tested with a variety of time intervals and controls
- Overall respondent-level analysis indicates significant, but very small, negative impact of decay on 4 brand metrics at 95% level
  - Aided Brand Awareness
  - Aided Online Advertising Awareness
  - Brand Favorability
  - Purchase Intent
- Positive impact of frequency on Aided Brand Awareness, Brand Favorability, and Purchase Intent at 95% level
- No significant interaction effects between time since last exposure and frequency were observed

Findings – Respondent-level

- Differences were observed across industry verticals for some metrics
  - Pharmaceutical category was the only vertical to exhibit significant decay in Online Ad Awareness at the individual respondent level
  - No significant effect of online ad exposure frequency was seen at the individual level within Automobile category when controlling for recency and demographic differences

Decay of Aided Brand Awareness



• Relationship between brand score and time since last exposure is not monotonic and appears to be affected by intervening and other individual factors; other variables (such as demographics) played a significant role

Results: Aggregated Data

Distribution of cases

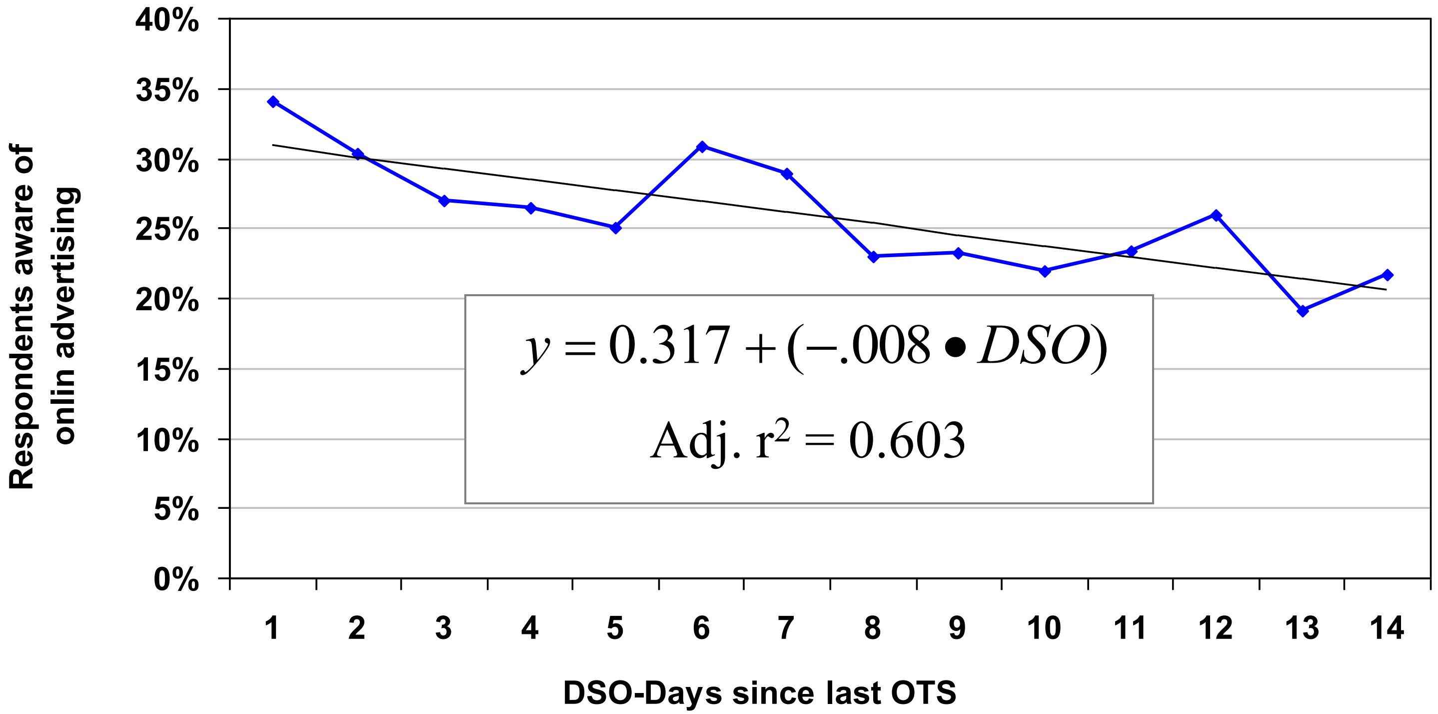
Time differential (days)	Cases		Time differential (days)	Cases
1	27,039		8	76
2	253		9	59
3	160		10	35
4	103		11	32
5	85		12	28
6	83		13	21
7	85		14	23

Findings – Aggregated data

- Only one of five brand metrics measured exhibited evidence of decay
  - Aided Online Advertising Awareness
  - "Days since last OTS" contributed a significant coefficient at 95% confidence

- Aided Brand Awareness, Message Association, Brand Favorability, and Purchase Intent were not significantly predicted by recency of OTS
  - No clear evidence of decay in this data set

Decay of Aided Ad Awareness



- For every day since last OTS, aggregate aided online advertising awareness decreases 0.8 percentage points.

Conclusions/Implications

- Both analytical methods suggest effects of decay on advertising effectiveness are limited

- Disaggregated analysis using logistic regression indicates that recency and frequency alone offers poor predictive ability; Aggregated data also shows limited effects

- Because rate of decay is not dramatic, decay may not significantly impact fairness of Internet-based ad effectiveness measurement

Recommendation for future research

- Primary data can be applied to any brand for situation-specific insights
  - Specifically recruit respondents at various points since last OTS
  - Expectation is that any number of variables influence rate of and magnitude of decay

- Study could be replicated within cross-media research study framework

- Once decay dynamics are known, media planner should create scenarios to inform scheduling

Thank you!

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